

WASP-135b

R-0129

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_241009 · created 2026-07-16T06:07:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460592.651 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.052
Transit depth (Rp/Rs)^2	3.07 +/- 1.81 [%]
Orbital Inclination (inc)	86.88 +/- 2.83
Airmass coefficient 1 (a1)	0.77 +/- 0.35
Airmass coefficient 2 (a2)	0.255 +/- 0.03
Scatter in the residuals of the lightcurve fit is	54.75 %
Best Comparison Star	#1 - [423, 384]
Optimal Aperture	2.22
Optimal Annulus	13.61
Transit Duration (day)	0.0881 +/- 0.0095

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.175 ± 0.052 vs 0.139 (deviation 0.15σ)
Duration fit vs published	0.0881 d vs 0.0687 d (deviation 0.45σ)
Mid-time O–C	-20.9 min
Depth SNR	0.7
Residual scatter	54.75 %

Light curve

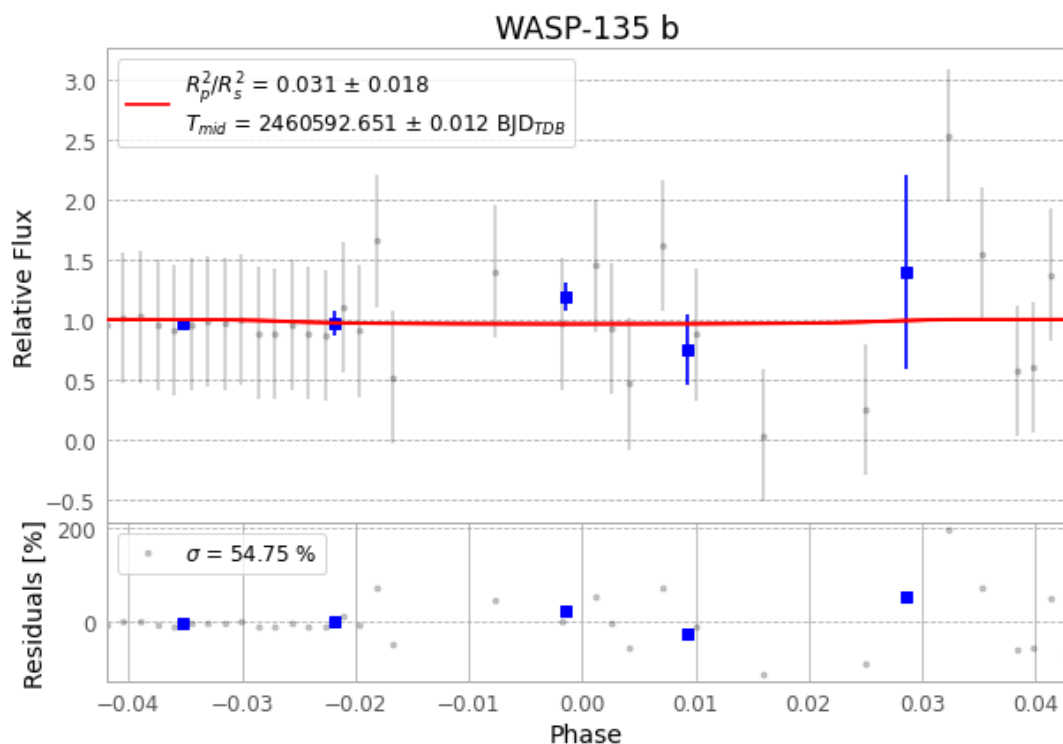


Figure 1 — FinalLightCurve_WASP-135 b_2024-10-08.png (SHA-256 0d3d31c84dc90921...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [325, 256], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d545361a513c48a8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

58 FITS frames (38 MB), WASP-135241009021216.FITS → WASP-135241009050316.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-241009.log (a7ac6645eb48...), FinalLightCurve_WASP-135 b_2024-10-08.png (0d3d31c84dc9...), FinalLightCurve_WASP-135 b_2024-10-08.csv (d3fcefd26e0b...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:08Z.